Lecture 1: Course Overview and Introduction

DSAA 6000J Foundations of Language Agents
2026 Spring

Zhijiang Guo

Instructor

- Bio: Zhijiang Guo (郭志江), Assistant Professor, DSA
- Research Interests: Large Language Models
- Email: zhijiangguo@hkust-gz.edu.cn

Hi! 👋 My name is Zhijiang Guo (郭志江). I am an Assistant Professor at the DSA Thrust, HKUST (GZ). I am also an Affiliated Assistant Professor of HKUST. I was a Senior Researcher at Huawei Noah's Ark Lab. Before that, I was a Postdoc at the Department of Computer Science and Technology at the University of Cambridge, working with Prof. Andreas Vlachos. I am also a member of Trinity College.

I earned my PhD in Computer Science from SUTD in 2020, under the supervision of Prof. Wei Lu. I was a visiting student at the University of Edinburgh from 2019-2020, working with Prof. Shay Cohen and Prof. Giorgio Satta on Structured Prediction. I also gained valuable insights from Prof. Zhiyang Teng. Before that, I was an undergraduate student at Sun Yat-sen University.

White Cliffs of Dover, England

I actively seek strong and motivated students to join our group! Feel free to email me if you are interested. More details can be found in Prospective Students and Visitors/中文版.

I am interested in natural language processing and machine learning, with a particular focus on large language models (LLMs). My research explores fundamental research questions about the **knowledge** and **reasoning** of LLMs, examining how these systems understand and process information.

Lab Introduction

- **LAnguage Reasoning and Knowledge Lab (LARK Lab)**
- Homepage: https://cartus.github.io/
- Bilibili (LLM Seminars): https://space.bilibili.com/3546570293250832?spm_id_from=333.337.search-card.all.click

Graduate Teaching Assistant

- Bio: Weile Luo (罗炜乐) is a Ph.D. student in Data Science at HKUST(GZ), supervised by Prof. Xiaowen Chu and Prof. Hongyuan Liu, with a research focus on GPU computing, LLM systems, and High-Performance Computing (HPC).
- Email: w.luo@connect.hkust-gz.edu.cn

Graduate Teaching Assistant

- Bio: Zherui Yang (杨哲锐) is a first-year PhD student in the AI Thrust, supervised by Prof. Hao Liu. His research interests focus on data science and large language models, with a particular emphasis on LLM-based agents and agentic reinforcement learning.
- Email: zyang582@connect.hkust-gz.edu.cn

Coding Lab

- Unfortunately, this course is primarily lecture-based, with no TA-led coding labs to guide you through the actual coding. 😔
- You'll have to figure that out on your own.

GitHub Hugging Face CLAUDE CODE

Topics covered by the lectures (Tentative)

- Week 1 Course Overview and Introduction
- Week 2 Large Language Model
- Week 3 Test-Time Scaling
- Week 4 LLM Reasoning
- Week 5 Memory and Planning
- Week 6 Trainingt-time Scaling
- Week 7 Coding Agents
- Week 8 Tool Agents
- Week 9 Multimodal Agents
- Week 10 Math Agents
- Week 11 Deep Research Agents
- Week 12 Open-Ended Evolution of Language Agents
- Week 13 Trustworthy and Safe Agentic AI

Invited talks given by industrial researchers

- **Research Scientist from MSRA (Beijing),** who won the NeurIPS Best Paper Runner-Up Award, will talk about training LLMs
- **Research Scientist from Huawei (Hong Kong)**, who won the ACL Outstanding Paper Award, will talk about tool agents.
- **More to come:** still reaching out more researchers from Meta, Tencent, etc.

Course Structure

- **Session 1:** Lecture (45-50 mins); Break
- **Session 2:** Lecture (45-50 mins); Break
- **Session 3:** Group Discussion and Presentation;

Course Assessment

Task	Contribution (%)
Project Paper	50%
Project Proposal	10%
Project Video	20%
In-Class Presentation	20%

- Project-based Learning: do a project and present your project.
- Outcome-based Assessment: if you master how to research a real-world problem with language agents and how to present the results to people, you will get a good grade and a good start of academic training.

Project & Presentation

- **Project Proposal:** Submit a one-page pdf (the NeurIPS formatting). Get an early feedback for your project. Provide an outline of your plan for the project and questions you will investigate / analysis you'll conduct in the course of it.
- Due: March 20th, 11:59 PM.

- **Project Paper:** Submit a paper, with 4-8 pages main text and additional appendices.
- Produce a conference-level paper that showcases a comprehensive understanding of the subject matter, significant innovation, meticulous attention to detail, and publishable quality, advancing the field.
- Due: May 15th, 11:59 PM.

Project & Presentation

- **Project Video:** Submit a oral presentation video (8-10 min) and the presentation slides. Recording an oral presentation video for your project. It would be better to record your talking face with the video, suggested by top conferences oral presentation instructions.
- Due: May 15th, 11:59 PM.

- **In-Class Presentation:** Actively involve in the group discussion and presentation
- Due: Every Session 3 you can get extra bonas for presenting, no advance preparation is required.

Projects related to your outside work

There is **no requirement to start a brand-new project specifically for this course.** Since this is a graduate-level course, I assume many of you are already working on active research.

The goal is to leverage the course material to **enhance your current academic trajectory,** ensuring the work remains uniquely your own.

- Integration: the primary requirement is that you apply the concepts and techniques of Language Agents discussed in this class **to your existing, ongoing research projects.**
- Authorship: You may not submit work for which you are not the primary (first) author.
- Collaboration Transparency: If the project involves contributions from multiple authors, you must include a brief clarification of your primary contribution to the work upon submission.

Minimum requirement

I understand that 8 pages conference-level paper might be challenging for some students. In this case, you can choose 4 pages workshop-level paper based on your situations:

- **Workshop-Level Paper (4 Pages):** This meets the minimum requirement for the course and is perfect for concise, focused projects.
- **Main Conference-Level Paper (8 Pages):** This is intended for more comprehensive research and detailed analysis.

In-class presentation

Attendance will not be formally tracked for this course. However, everyone is guaranteed **at least 80%** of the total points available in the In-Class Presentation category (16 points).

To boost your grade further, you can earn 2 bonus points for every presentation you give. These bonus points are capped at 12 points total, meaning you can earn up to 8 extra credit points if you choose to present multiple times.

- Bonus Opportunity: +2 points per presentation.
- Bonus Cap: Maximum of 8 additional points (totaling 12 points in the bonus pool).

Grading Scheme

- Excellent Performance: [100,85]; (A+/A/A-)
- Good Performance: (85,75]; (B+/B)
- Satisfactory Performance: (75,60];(B-/C+/C)
- Fail: (60,0].(F)
- Subgrades (e.g., A-, B+, etc.) will be assigned for every 5 points within these ranges.

Intended Learning Outcomes (ILOs)

By the end of this course, students should be able to:

- 1. Understand the foundation of language agents: models and different tasks.
- 2. Train and use language agents.
- 3. Create/design language agents with other researchers.
- 4. Explain/present their results to others.

What We Expect and What You Earn

- You should not seek assistance from any outside supervisors. You should be the first author for your project, with the assistance only from the instructor/GTAs.

- It's wonderful, if this project paper into a publication. This is very likely, if you get a good grade for your project.

- A start of academic training for junior PG students is also a big first-mover advantage, at least in AI community. We accept it and enjoy it. We hope what you earn in this course can be useful for your future research.

AI and Human Assistant Policy

- **Human Assistant: while discussion is encouraged, copy or plagiarization is strictly prohibited.**
- **AI Policy: treat AI assistants as human assistants.**
- The policy for using AI assistants is identical to the policy for using human assistants. Imagine the AI as a human and apply the same norm as you would with a human.
- This is an AI class and you should try out all the latest AI assistants. It's very important to play with them to learn what they can do and what they can't do. That's a part of the content of this course.
- Just like you can come to office hours and ask a human questions (about the lecture material, clarifications about assignments, tips for getting started, etc), you are very welcome to do the same with AI assistants. But, just like you are not allowed to ask an expert friend to do your homework for you, you also should not ask an expert AI.

Assignment and Exam

- We do not have any assignments, please read the related paper and projects if you are interested in this topic.
- We do not have any exam or quiz, please focus on your research project.

Situations where you should not take this course

- If you prefer exam-based evaluation, this course is not suitable. This course is 100% project-based. I often receive complaints that the project is “**extremely time-consuming”**, as it is an individual project rather than a group project.

Situations where you should not take this course

- This course may be better suited for students **with prior research experience.** Since the research project is due in May, the timeline is quite tight. Additionally, we do not have coding labs, so students without relevant background or experience may receive unsatisfactory grades.

Situations where you should not take this course

- **Every class session includes group discussion and presentation**. Although no advance preparation is required, some students *may prefer three-hour lectures or wish to skip classes frequently*. Please note that these activities serve as attendance checks.

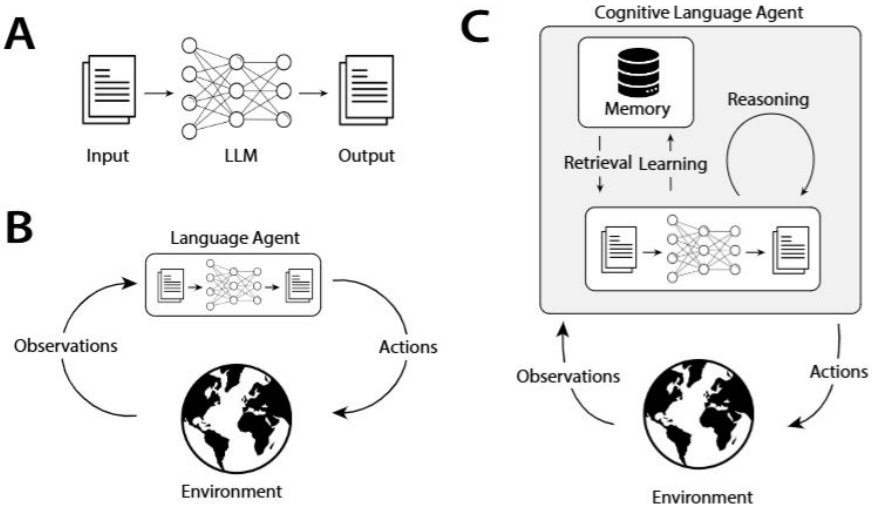
Situations where you should not take this course

- **Other reasons:**
  - Instructor’s accent is weird;
  - Dont like invited talks by external speakers;
  - Procrastinator（拖延症）
  - …

Today's Lesson

- Introducing the foundation of language agents: language models

Different uses of LLMs



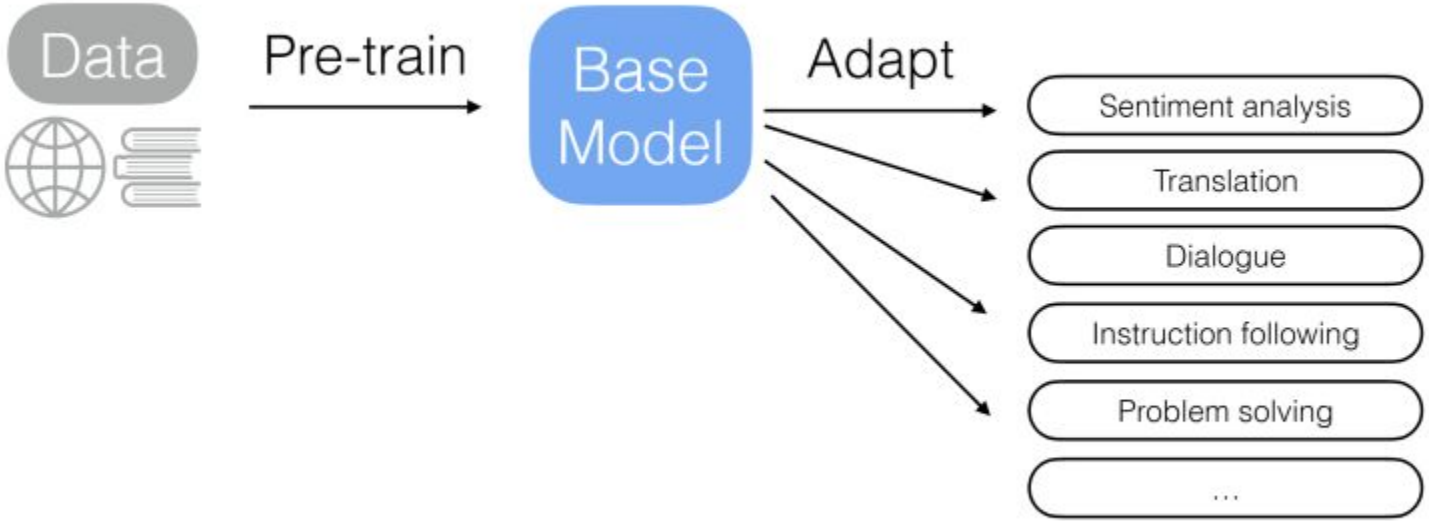
- A: In NLP, an LLM takes text as input and outputs text.
- B: Language agents place the LLM in a direct feedback loop with the environment by transforming observations into text and using the LLM to choose actions.
- C: Cognitive language agents additionally use the LLM to manage the agent's internal state via processes such as learning and reasoning

Cognitive Architectures for Language Agents. TMLR 2024

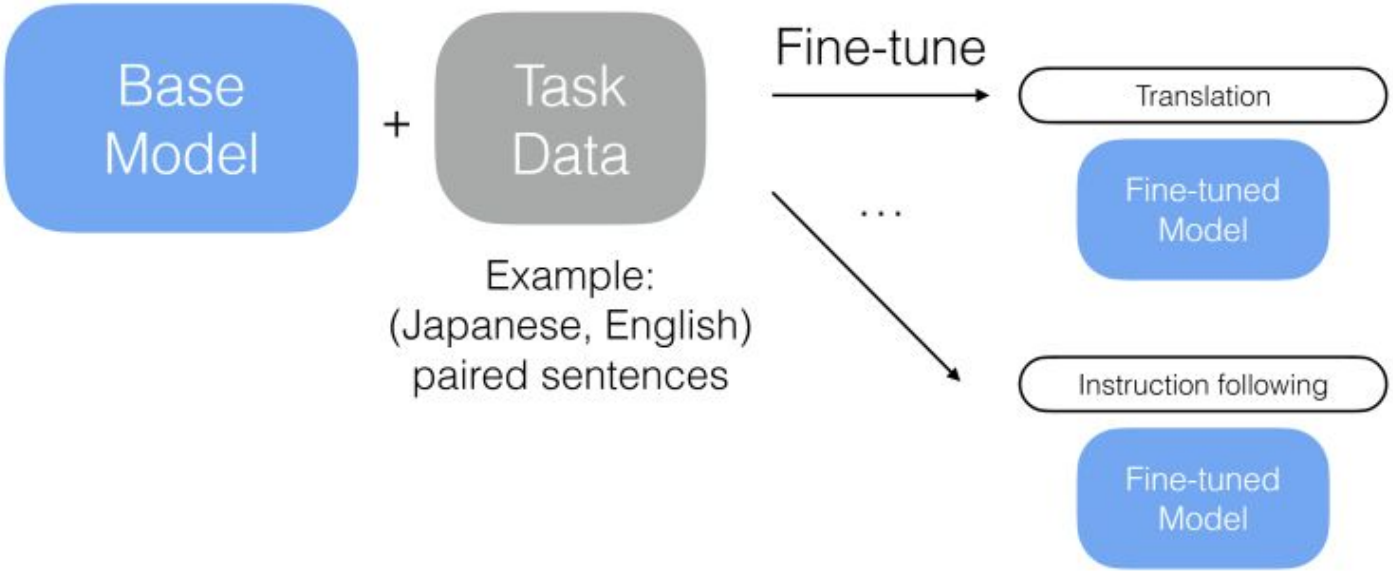
Language Models

- Previously:
  - Train model from scratch
  - 1 model for 1 task
- Now:
  - Pretrain a single model, adapt it to many tasks.

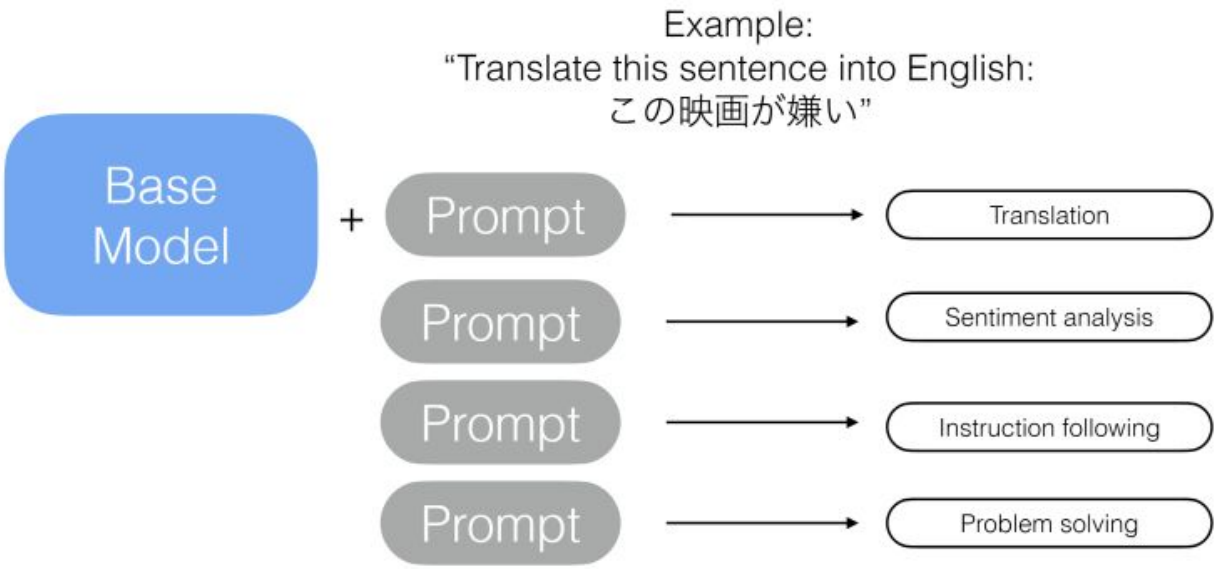
Basic Idea


Data
Pre-train
Base Model
Adapt
Sentiment analysis
Translation
Dialogue
Instruction following
Problem solving
...

Adaptation: fine-tune



Adaptation: prompting



Why pre-train?

Transfer learning: take “knowledge” from one task and apply it to another task

- Less task data: use less data to reach a given level of performance
- Better task performance: reach higher performance than training from scratch
- One model, multiple tasks: convenient, amortizes cost, a starting point for many uses, . . .

Major factors

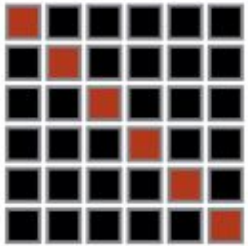
- Pre-trained models have names like BERT, Llama2, Deepseek-V3, Claude-4.5, GPT-5. . .
- Each model is influenced by 4 major factors:
  - Model: The underlying neural network architecture
  - Training objective: The objective used to train
  - Data: the data used to train the model
  - Hyperparameters: e.g. learning rate, batch size

Which model

- Usually Transformer, although the details vary
- Size: bigger usually has better task performance within a given model family
- Model details can vary (or are underspecified)

Which Objective?

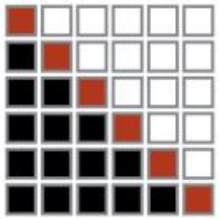
$$P(X) \neq \prod_{i=1}^{|X|} P(x_i | x_{\neq i})$$



Masked language modeling: used more for fine-tuning (e.g. BERT)

Which Objective?

$$P(X) = \prod_{i=1}^{|X|} P(x_i | x_1, \ldots, x_{i-1})$$



Auto-regressive language modeling: used for finetuning, prompting. Extremely popular in 2025 (e.g. DeepSeek V3)

Which Data?

Data is extremely important, common sources

- (2018) Books corpus - a large corpus of books
- (2018-) Wikipedia
- Common crawl - data from the whole internet . . .

Which hyper-parameters?

- Reported for some models
- We'll go over strategies that are used to select some of them

GPT-4 Technical Report

OpenAI*

Abstract

we used python

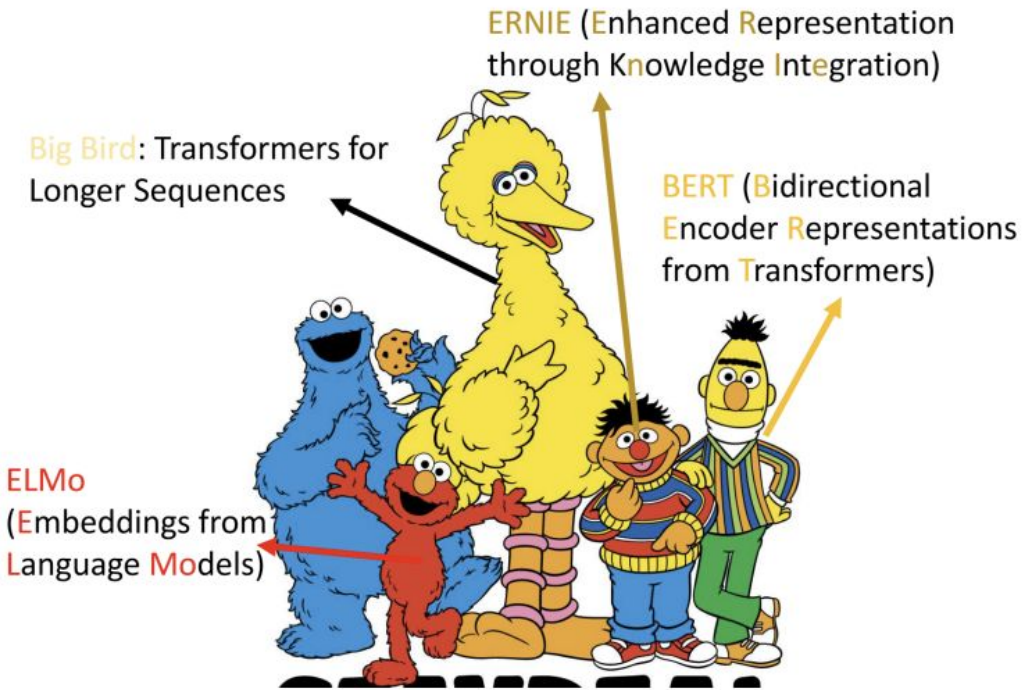
Today's lecture

- Objectives
  - Masked language modeling
  - Autoregressive language modeling
- Data: sources, quality, and quantity
- Thinking about pretraining
  - Tokens, model size, compute
  - Scaling laws

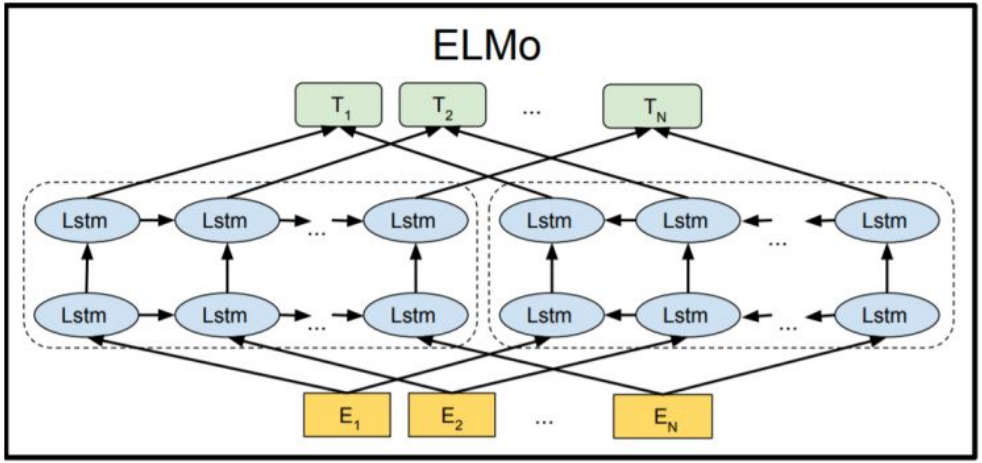
Today's lecture

- Objectives
  - Masked language modeling
  - Autoregressive language modeling

Masked language modeling



Masked language modeling (ELMo)



- The heart of ELMo is a large bi-LSTM (94M), two independent LSTMs process the input sequence in opposite directions: forward and backward.
- The outputs of these two LSTMs are concatenated to form a single vector representation for each word in the input sequence.

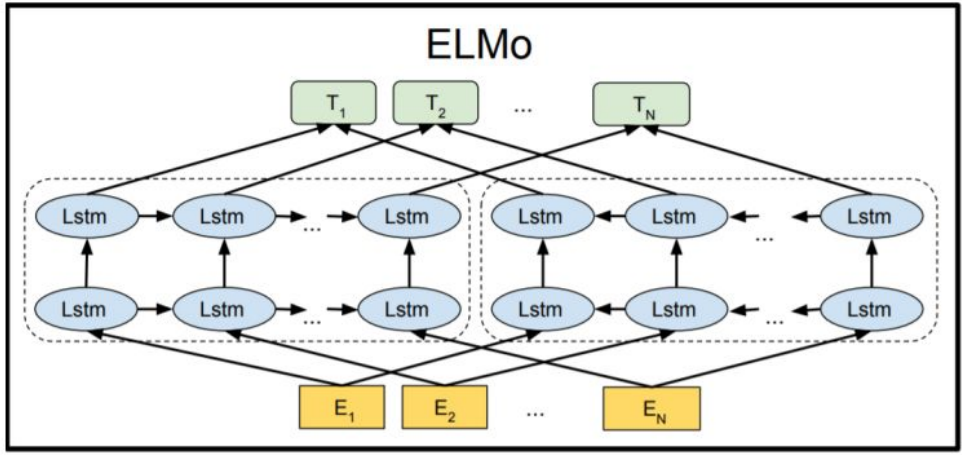
Deep contextualized word representations. NAACL 2018

Word Embedding

Token Index	Token Embedding
0	[0.235, -1.256, 3.513, ..., -0.187]
1	[1.291, -2.012, 0.624, ..., -1.291]
2	[0.536, 0.012, -0.024, ..., 2.345]
...	...
Vocab Size |V|	[0.131, 2.102, 0.935, ..., -0.125]

A look-up table that converts token index to continuous vectors (static: a single vector regardless of context.)

Masked language modeling (ELMo)



- Concatenation of forward and backward LSTM outputs represents the contextual information surrounding each word.
- Resulting representations are significantly more complex and informative than static word embeddings, allowing for better performance in various NLP tasks.

Glove vs. ELMo

	Source	Nearest Neighbors
GloVe	play	playing, game, games, played, players, plays, player, Play, football, multiplayer
biLM	Chico Ruiz made a spectacular play on Alusik 's grounder {...}	Kieffer , the only junior in the group , was commended for his ability to hit in the clutch , as well as his all-round excellent play .
	Olivia De Havilland signed to do a Broadway play for Garson {...}	{...} they were actors who had been handed fat roles in a successful play , and had talent enough to fill the roles competently , with nice understatement .

Table 4: Nearest neighbors to "play" using GloVe and the context embeddings from a biLM.

Embedding Type

- GloVe: Produces static embeddings—each word has a single vector regardless of context.
- ELMo: Produces contextualized embeddings—a word's vector changes based on its surrounding context.

Glove vs. ELMo

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Context Handling

- GloVe: Cannot distinguish between different meanings of a word (e.g., "bank" as financial vs. river).
- ELMo: Captures context using a bidirectional LSTM, enabling it to resolve polysemy

Masked language modeling

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. NAACL 2019

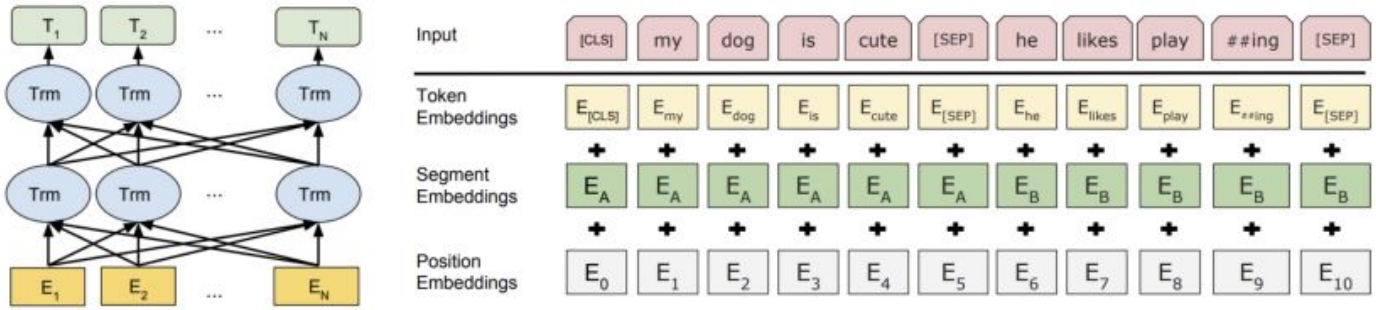
Masked language modeling

System	MNLI-(m/mm)	QQP	QNLI	SST-2	CoLA	STS-B	MRPC	RTE	Average
	392k	363k	108k	67k	8.5k	5.7k	3.5k	2.5k	-
Pre-OpenAI SOTA	80.6/80.1	66.1	82.3	93.2	35.0	81.0	86.0	61.7	74.0
BiLSTM+ELMo+Attn	76.4/76.1	64.8	79.8	90.4	36.0	73.3	84.9	56.8	71.0
OpenAI GPT	82.1/81.4	70.3	87.4	91.3	45.4	80.0	82.3	56.0	75.1
$\text{BERT}_{\text{BASE}}$	84.6/83.4	71.2	90.5	93.5	52.1	85.8	88.9	66.4	79.6
$\text{BERT}_{\text{LARGE}}$	**86.7/85.9**	**72.1**	**92.7**	**94.9**	**60.5**	**86.5**	**89.3**	**70.1**	**82.1**

Table 1: GLUE Test results, scored by the evaluation server (https://gluebenchmark.com/leaderboard). The number below each task denotes the number of training examples. The "Average" column is slightly different than the official GLUE score, since we exclude the problematic WNLI set.[8] BERT and OpenAI GPT are single-model, single task. F1 scores are reported for QQP and MRPC, Spearman correlations are reported for STS-B, and accuracy scores are reported for the other tasks. We exclude entries that use BERT as one of their components.

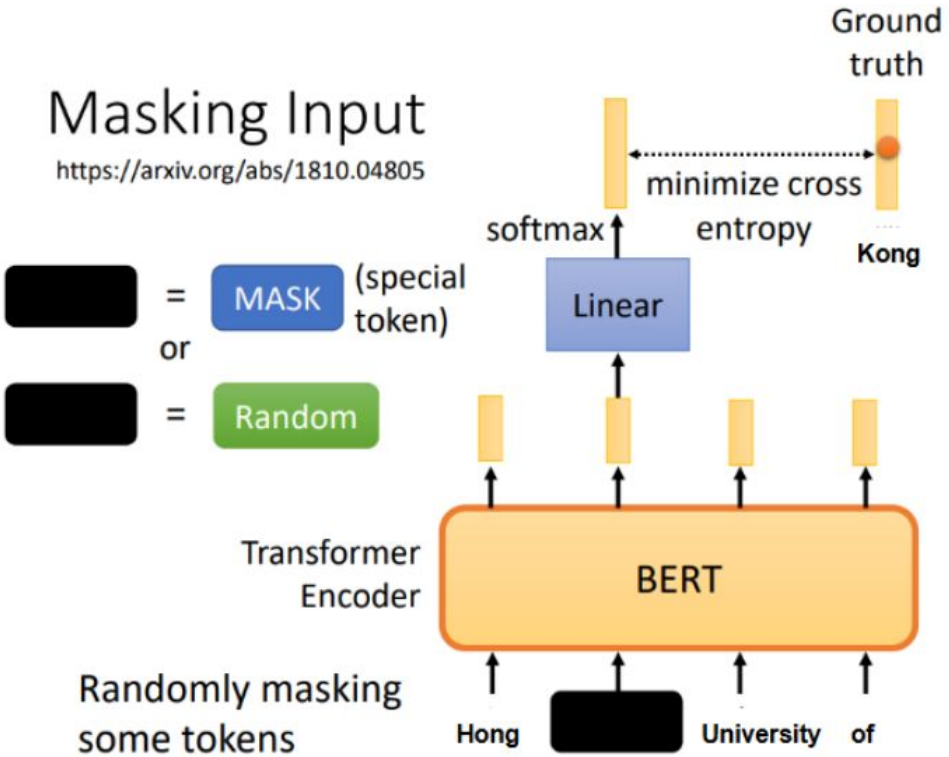
Significantly outperforms ELMo on all tasks.

Masked Language Modeling (BERT)

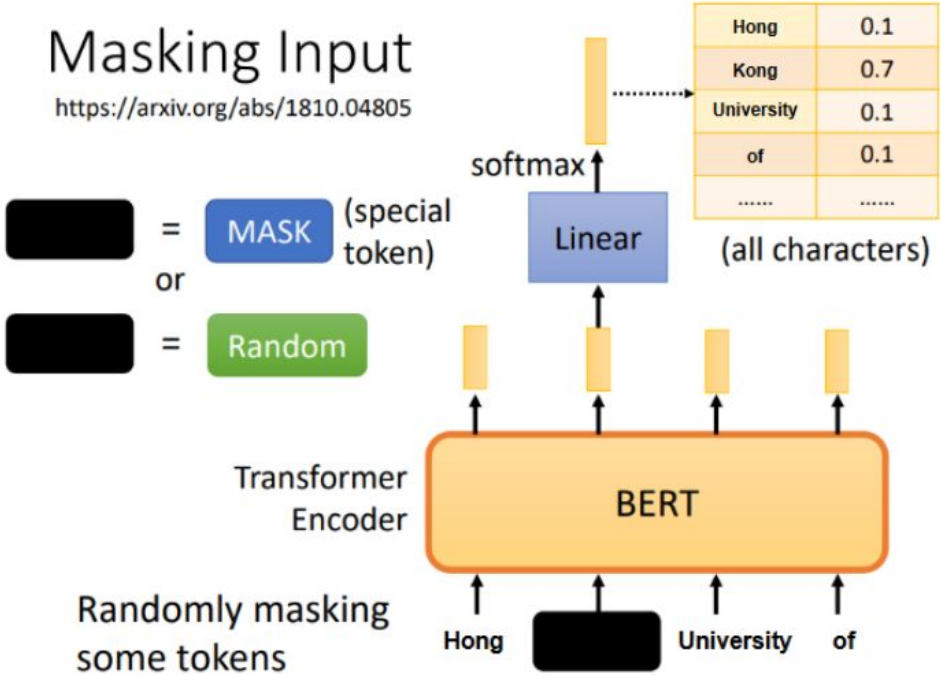


- Model: Multi-layer self-attention. Input sentence or pair, w/ [CLS] token, subword representation
- Objective: Masked word prediction + next sentence prediction
- Data: BooksCorpus + English Wikipedia

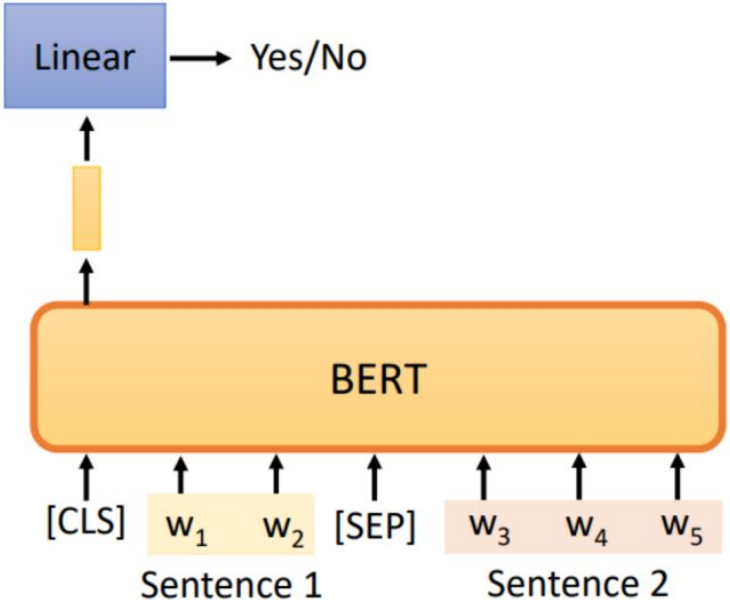
Masked Word Prediction



Masked Word Prediction



Next Sentence Prediction

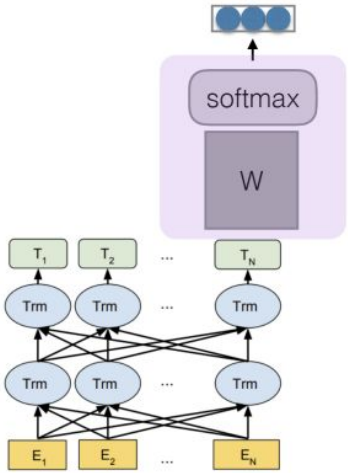


Masked Word Prediction

Predict a masked word

- 80%: substitute input word with [MASK]
- 10%: substitute input word with random word
- 10%: no change

Adapting a masked language model



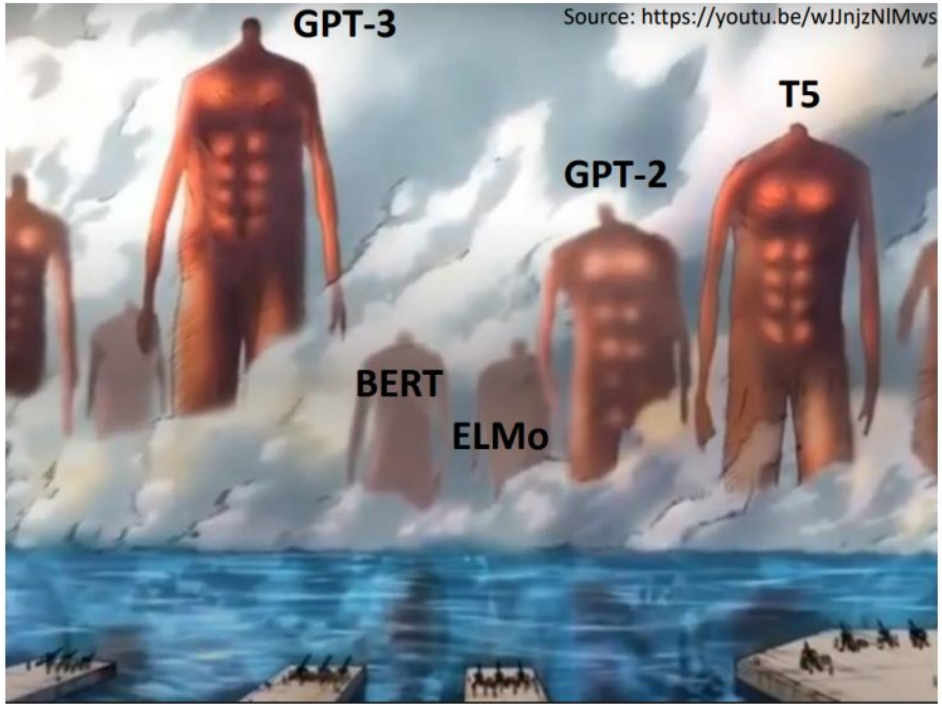
Add an output layer that maps a hidden vector to scores. Example:

- Data: (movie review, positive, neutral, negative)
- Initialize the model with BERT
- Train with cross-entropy loss

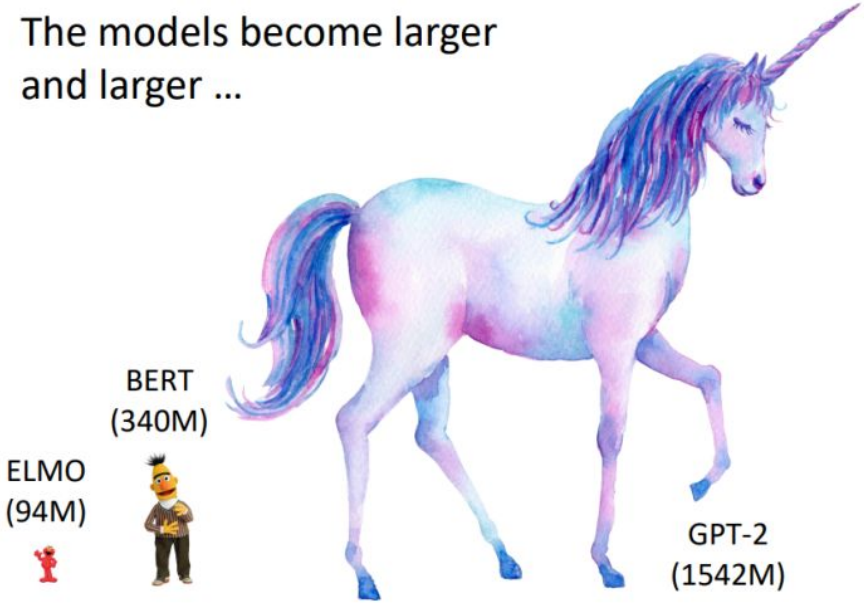
Today's lecture

- Objectives
  - Masked language modeling
  - **Autoregressive language modeling**

Autoregressive language modeling



Autoregressive language modeling



Language Models are Unsupervised Multitask Learners OpenAI Blog 2019

Autoregressive language modeling

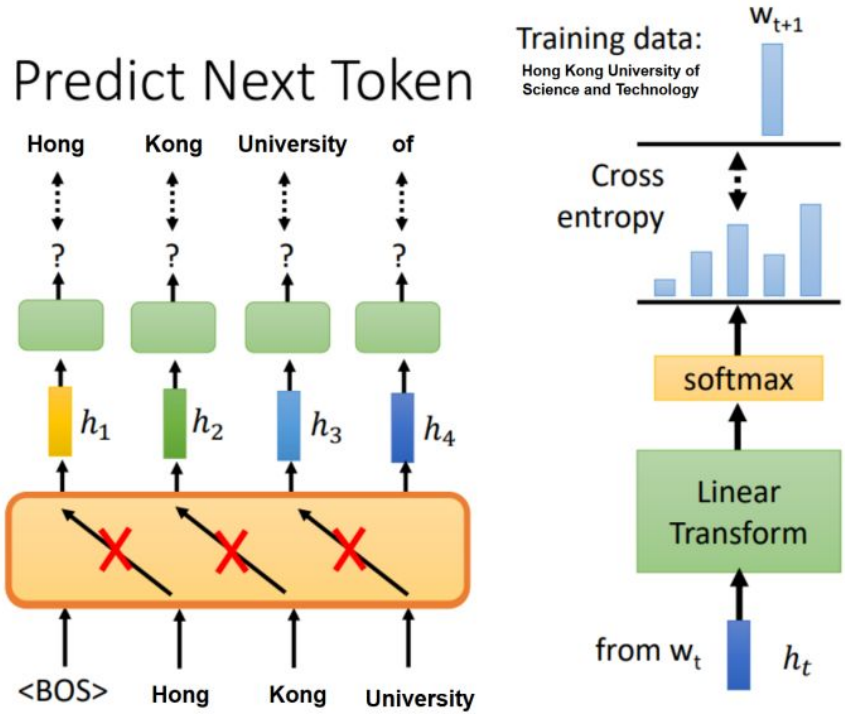
$$\min_{\theta} \sum_{x \in D_{train}} \sum_{t} -\log p_\theta(x_t \mid x_{<t})$$

$$\text{minimize}_\theta \quad \text{CrossEntropyLoss}(\theta, D_{train}, \phi)$$

Key inputs:

- θ: model architecture and size
- D_{train}: data (content, quantity)
- ϕ: hyper-parameters, e.g. learning rate, batch size

Autoregressive language modeling



Evaluating a mode

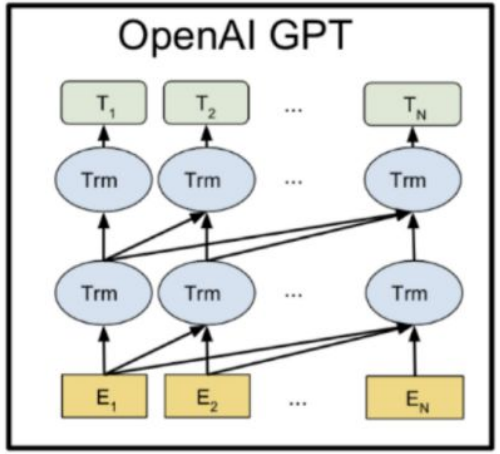
Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
Translate English to French:        ← task description
sea otter => loutre de mer          ← examples
peppermint => menthe poivrée
plush girafe => girafe peluche
cheese =>                           ← prompt
```

- Loss (training, validation, test) : Diagnose training trajectory, compare models in the same family
- Few-shot prompting
- Fine-tuning

Example: GPT-2

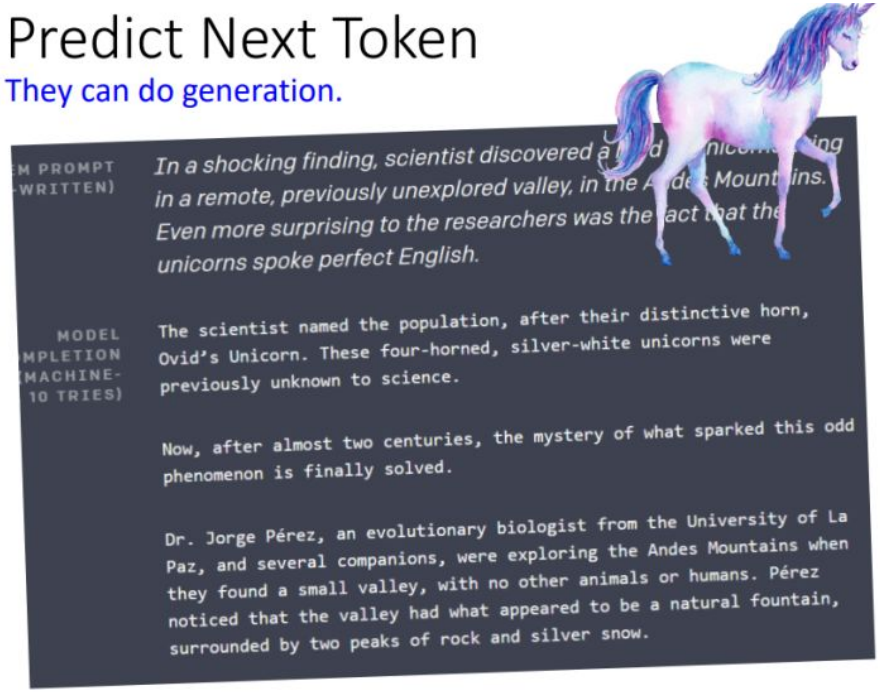


- Model: Transformer (1.5B)
- Data: WebText (millions of web pages)
- Results: Impressive results in generation of long form text, and zero shot task completion

Example: GPT-2

Predict Next Token

They can do generation.



- Impressive results in generation of long form text, and zero shot task completion

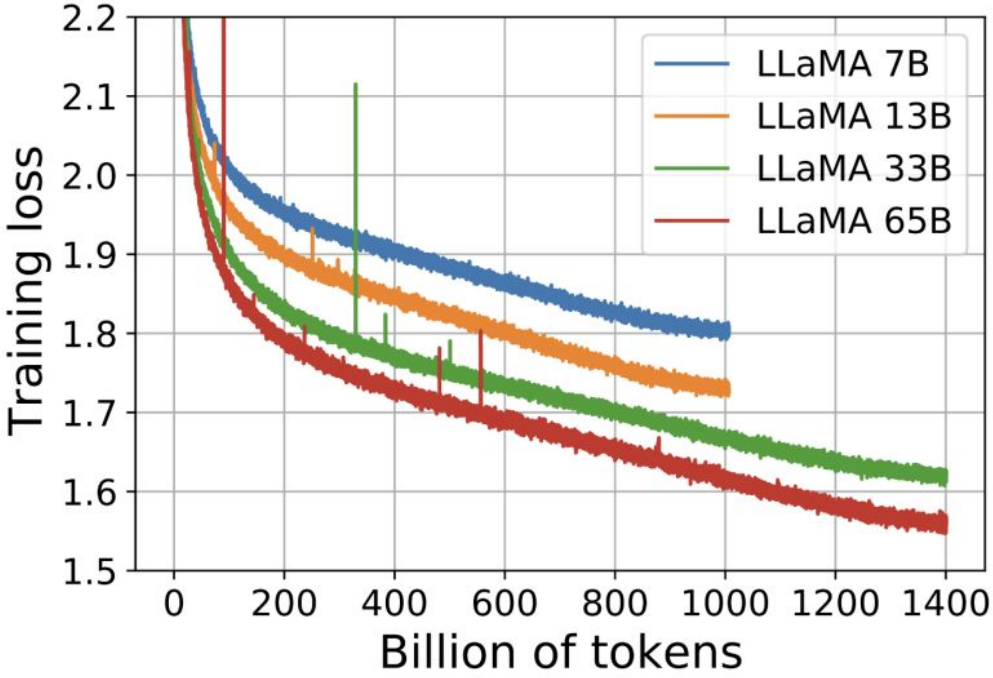
Example: Llama

Dataset	Sampling prop.	Epochs	Disk size
CommonCrawl	67.0%	1.10	3.3 TB
C4	15.0%	1.06	783 GB
Github	4.5%	0.64	328 GB
Wikipedia	4.5%	2.45	83 GB
Books	4.5%	2.23	85 GB
ArXiv	2.5%	1.06	92 GB
StackExchange	2.0%	1.03	78 GB

- Model: Transformer, 6.7B, 13B, 32B, 65B
- Data: 1.4 trillion tokens, sources:

LLaMA: Open and Efficient Foundation Language Models. Meta 2023

Llama: training los



Few-shot learning

```
Translate English to French:
sea otter => loutre de mer
peppermint => menthe poivrée
plush girafe => girafe peluche
cheese =>
```

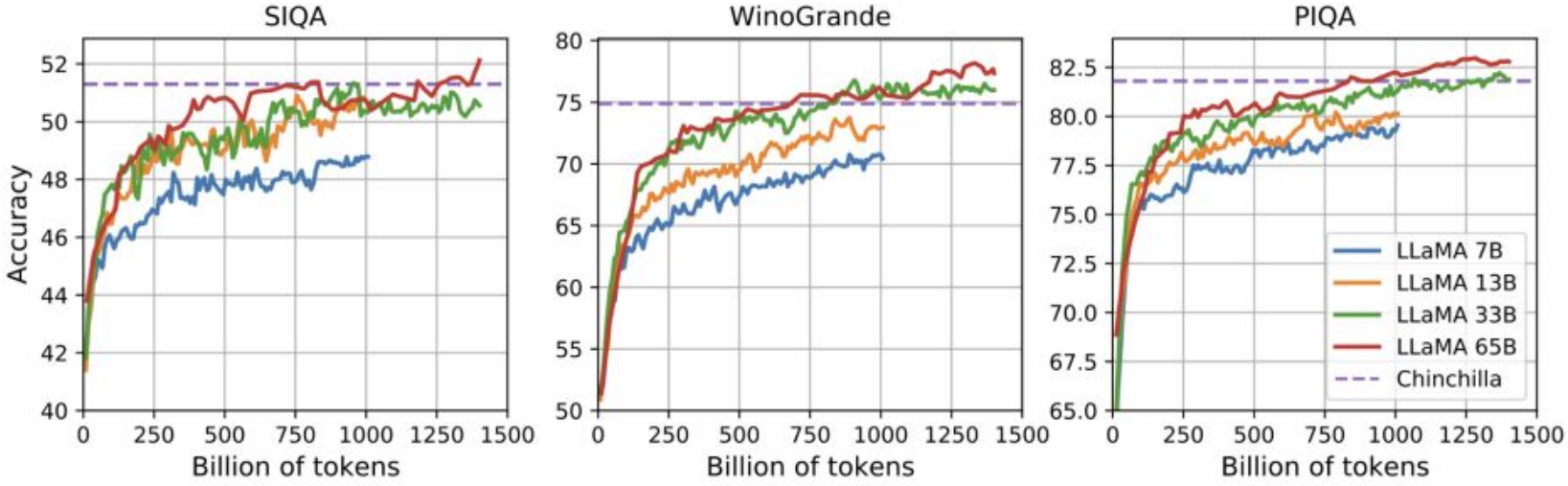
task description

examples

prompt

"Few-shot" Learning: (no gradient descent), also called in-context learning

Llama: few-shot performance trajectory



Practical tools: HuggingFace

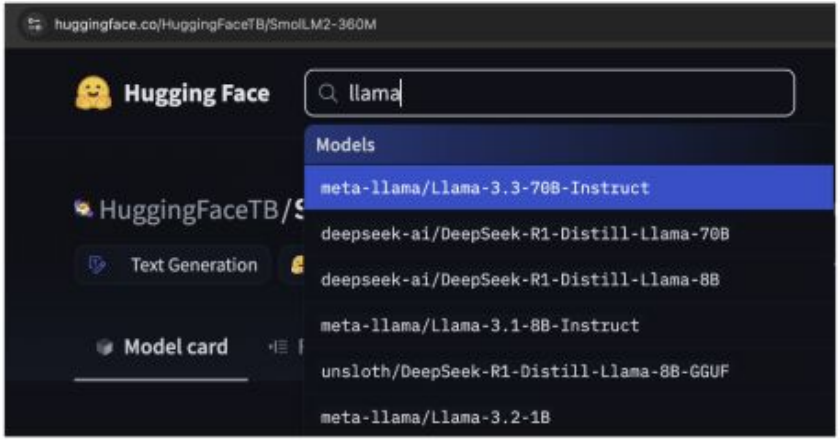
Load tokenizer and model

- Find models at https://huggingface.co/

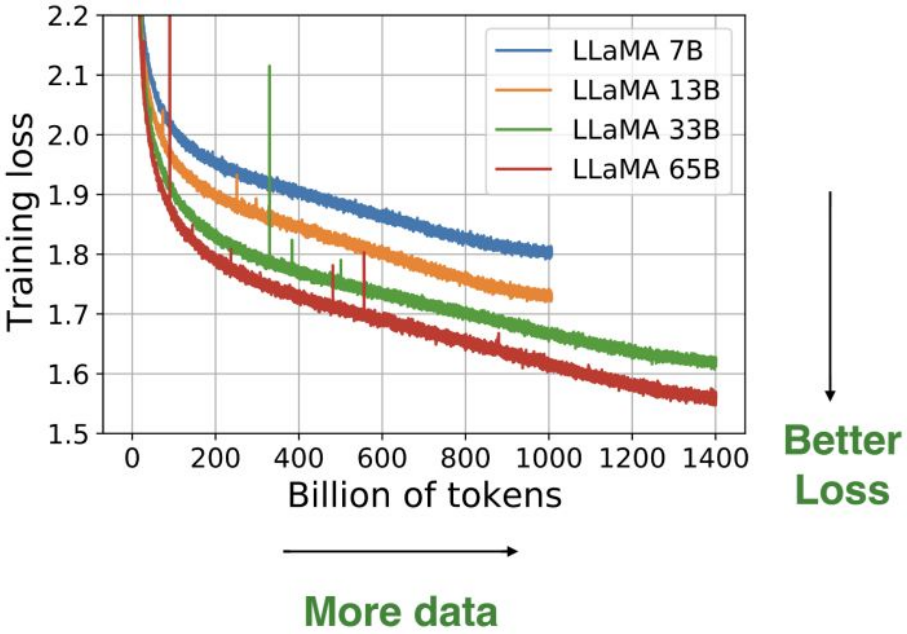
```
from transformers import AutoTokenizer, AutoModelForCausalLM

model = "HuggingFaceTB/SmolLM2-360M"

tokenizer = AutoTokenizer.from_pretrained(model)
model = AutoModelForCausalLM.from_pretrained(model)
```



Data: sources, quality, and quantity



Data factors

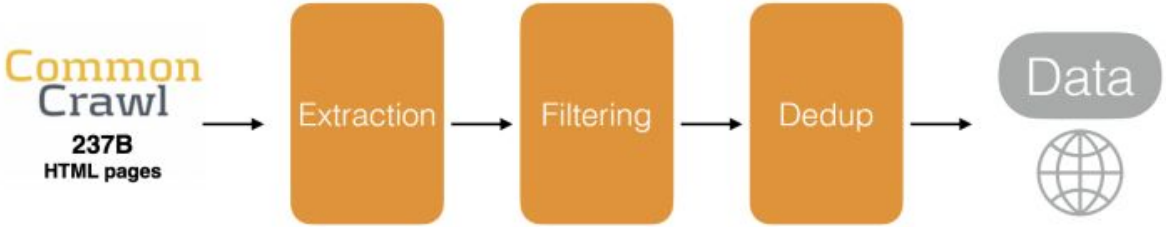
- Quantity: How much data do I have?
- Quality: Is it beneficial for training?
- Coverage: Does the data cover the domain(s) I care about, and in the right proportions?

Data quantities

	Tokens of training data
Llama 1	1.4 trillion
Llama 2	1.8 trillion
Llama 3	15 trillion
Deepseek 3	15 trillion

Wikipedia has less than 10 billion

Web data: common crawl



Large snapshots of web pages

- Extraction: HTML to text
- Filtering: filter out unwanted pages
- Deduplication: many duplicate web pages

Quality: Extraction

```
This paper concerns the quantity
<img src="https://s0.wp.com/
latex.php?latex=%7BM%28x%29..."
alt="{M(x)}" />, defined as the
length of the longest
subsequence of the numbers from
```

Image Equations

```
Suppose I have a smooth map
[tex]f\colon \mathbb{R}^3
\longrightarrow S^2[/tex]. If I
identify [tex]\mathbb{R}^3[/tex]
with [tex]U_S = S^3 - \
{(0,0,1)\}[/tex] via
stereographic projection
```

Delimited Math

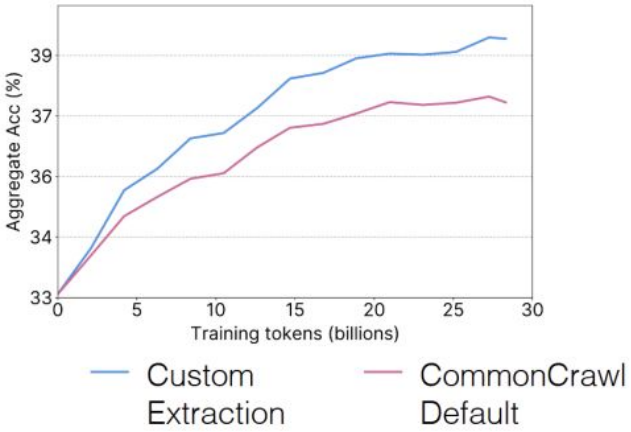
```
<math>
  <semantics>
    ...
    <annotation ...>
      {\displaystyle \mathrm {MA}
      ={\frac{f_{O}}{f_{E}}}}
    </annotation>
  </semantics>
</math>
```

Special Tags

Extraction: HTML to text

- Remove boilerplate
- Retain Latex, code, etc.

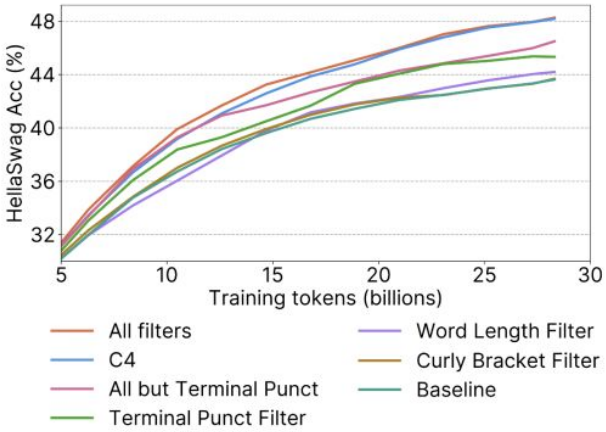
Quality: Extraction



Extraction: HTML to text

- Remove boilerplate
- Retain Latex, code, etc.

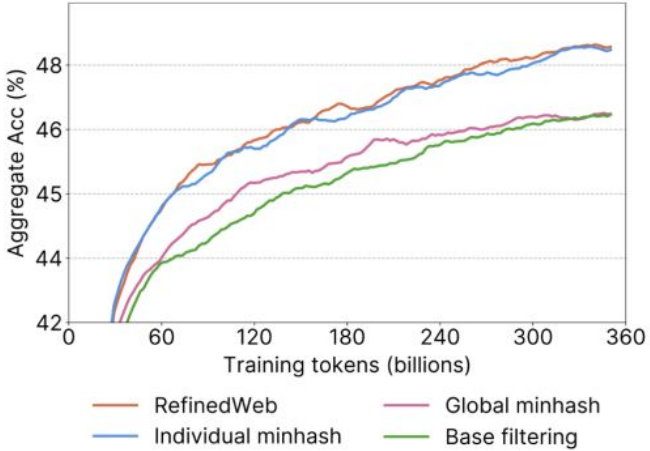
Quality: Filtering



Filter out unwanted text

- Language filter
- Repetitions
- Too many short lines

Quality: Deduplication



Remove duplicate content

- Fuzzy strategy: minhash
- Too much deduplication can be harmful

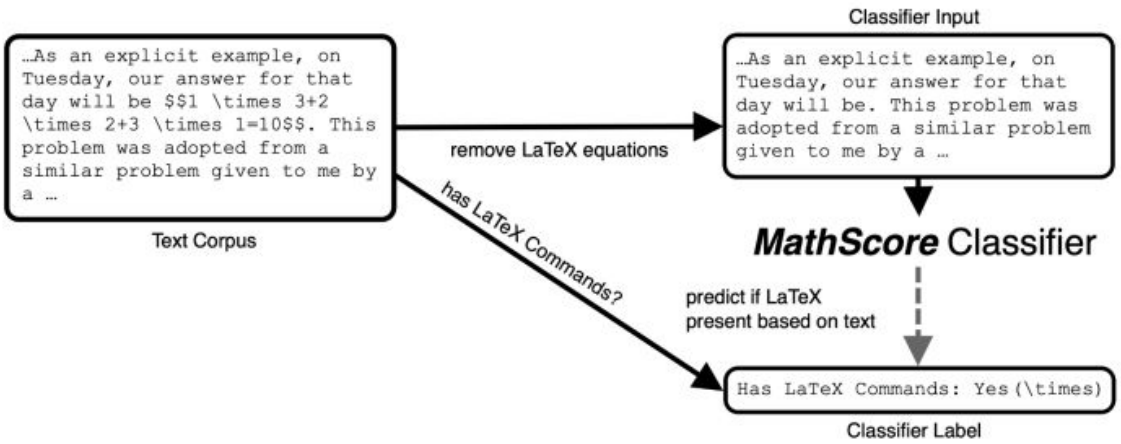
Coverage

- The data determines the data distribution
- And hence the model, $p_\theta \approx p_{data}$
- Web data ≠ educational data ≠ math data

Approach: classifier filtering

- Train a classifier to detect desired data
- Use it to filter out undesired data

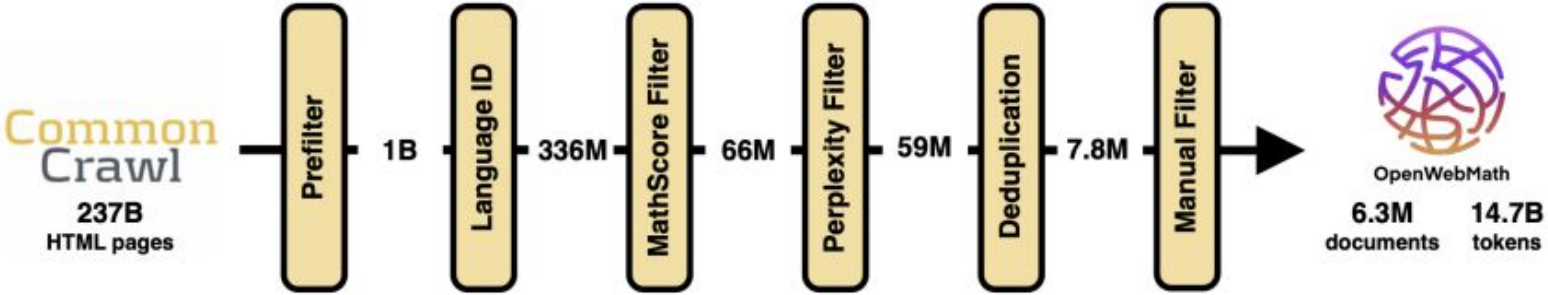
Approach: classifier filtering



Example: OpenWebMath

- MathScore classifier detects math content

Approach: classifier filtering



Example: OpenWebMath

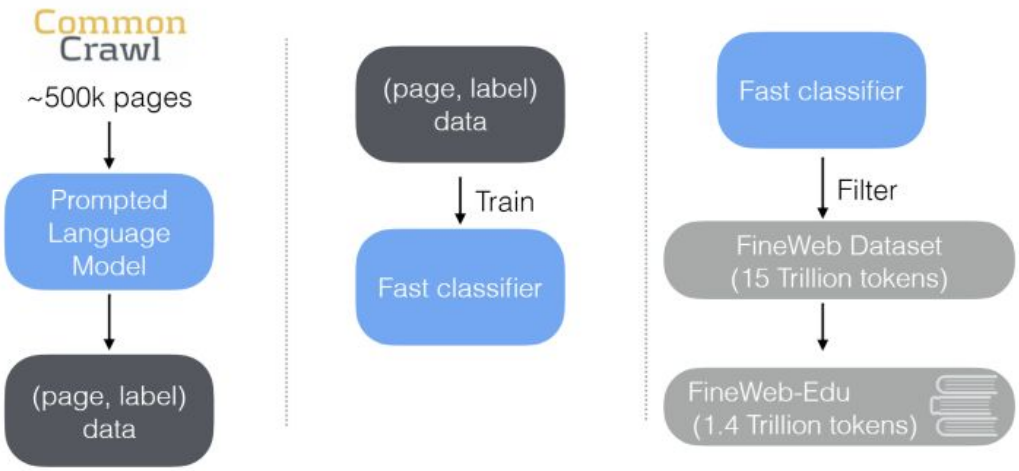
- MathScore classifier detects math content

Approach: classifier filtering

Domain	# Characters	% Characters
stackexchange.com	4,655,132,784	9.55%
nature.com	1,529,935,838	3.14%
wordpress.com	1,294,166,938	2.66%
physicsforums.com	1,160,137,919	2.38%
github.io	725,689,722	1.49%
zbmath.org	620,019,503	1.27%
wikipedia.org	618,024,754	1.27%
groundai.com	545,214,990	1.12%
blogspot.com	520,392,333	1.07%
mathoverflow.net	499,102,560	1.02%

Example: OpenWebMath

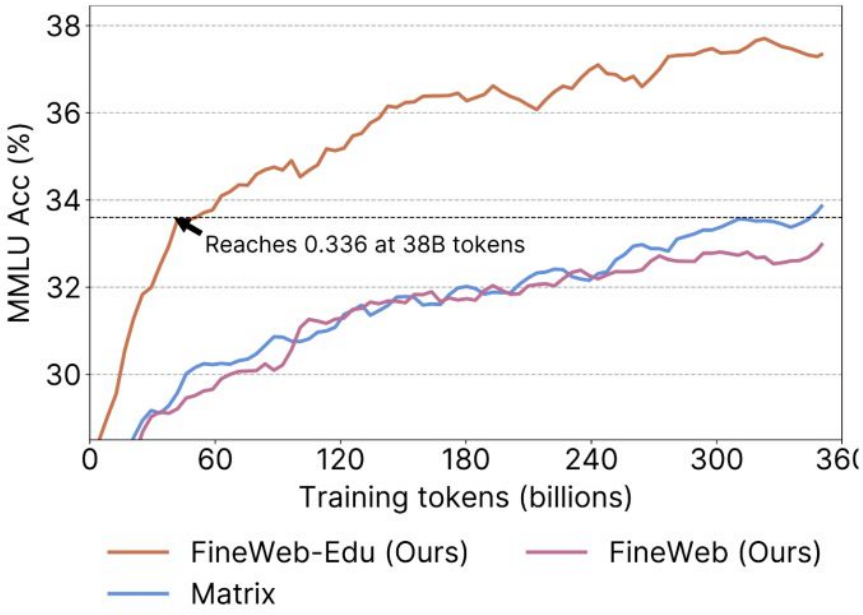
Approach: classifier filtering



Example: FineWeb-Edu

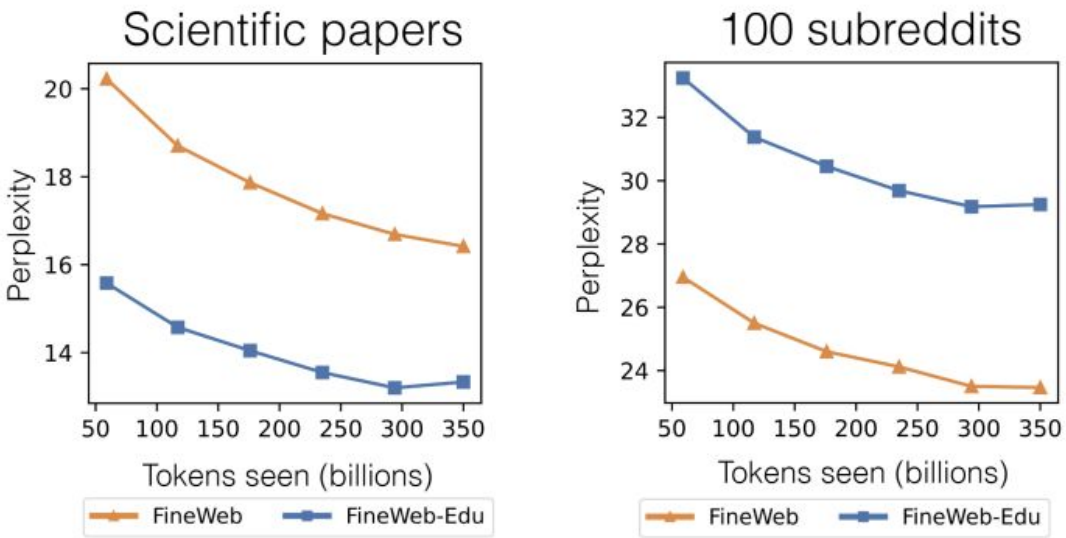
- Classifier to classify pages as "educational"

Approach: classifier filtering



Example: FineWeb-Edu

Approach: classifier filtering



Example: FineWeb-Edu

Mixtures

Source	Type	Tokens
	Pretraining ✦ OLMo 2 1124 Mix	
DCLM-Baseline	Web pages	3.71T
StarCoder filtered version from OLMoE Mix	Code	83.0B
peS2o from Dolma 1.7	Academic papers	58.6B
arXiv	STEM papers	20.8B
OpenWebMath	Math web pages	12.2B
Algebraic Stack	Math proofs code	11.8B
Wikipedia & Wikibooks from Dolma 1.7	Encyclopedic	3.7B
Total		**3.90T**

In practice, training data is a mixture of different sources

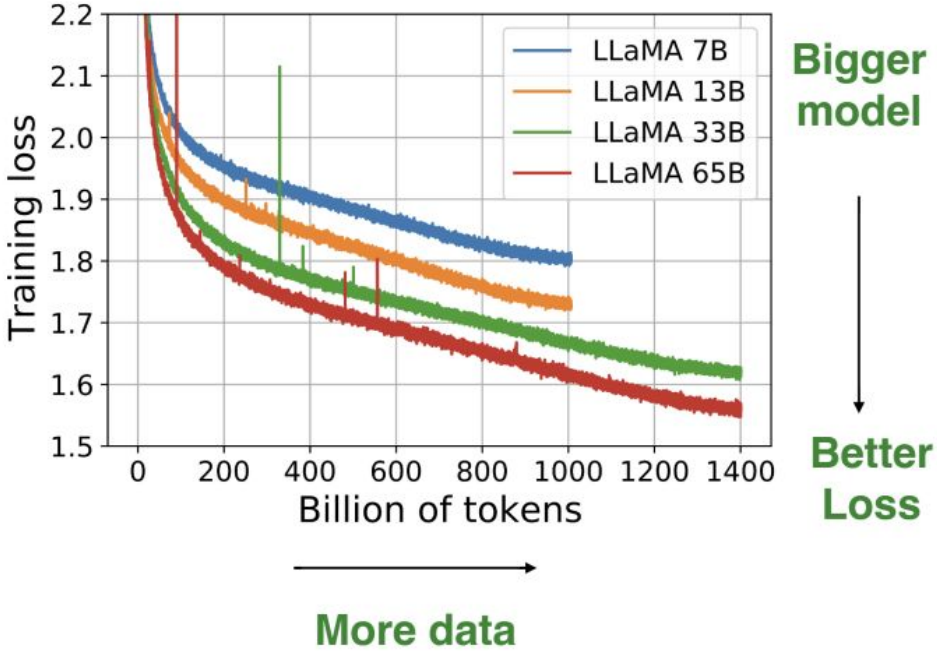
Recap

- Web data: large quantities of data
  - Extract, filter, deduplicate to improve quality
  - Filter to cover desired domain(s)
- Mix together web data and other sources to make a pre-training dataset

Recent examples

	Year	Domain	Tokens
FineWeb	2024	Web	15 trillion
RedPajama v2	2024	Web	30 trillion
Dolma	2024	Mix	3 trillion
OLMO2 Mix	2025	Mix	4 trillion
OpenWebMath	2023	Math web pages	15 billion
AlgebraicStack	2023	Math code	11 billion
FineWeb-Edu	2024	Educational (middle-school)	1.4 trillion

Thinking about pretraining



Pretraining and compute

Goal: get a better pretrained model by “adding more compute”

- *“The biggest lesson that can be read from 70 years of AI research is that general methods that leverage computation are ultimately the most effective, and by a large margin.”*

  - The Bitter Lesson, Richard Sutton 2019

Project Discussion

- What are your interested fields and future career directions?
- Name one promising (game-changing) AI application or idea in your field and discuss it with your group members.
- Start Your Project Now: Consider this as a hint for your project proposal.